

PROBLEM SET 3

Show **ALL WORK** to get full/partial credit. Begin each problem on a new page, and clearly label each part of the problem. (**Max Score:** 80)

1) **(10 pts)** Derive Eqs. (15.1.10) and (15.1.11). It might help to use

$$\mathbf{S}_1 \cdot \mathbf{S}_2 = S_{1z}S_{2z} + \frac{1}{2}(S_{1+}S_{2-} + S_{1-}S_{2+})$$

2) **(30 pts)** In addition to the Coulomb interaction, there exists another, called the *hyperfine interaction*, between the electron and proton in the hydrogen atom. The Hamiltonian describing this interaction, which is due to the magnetic moments of the two particles is,

$$H_{hf} = A \mathbf{S}_1 \cdot \mathbf{S}_2 \quad (A > 0)$$

(This formula assumes the orbital state of the electron is $|1,0,0\rangle$.) The total Hamiltonian is thus the Coulomb Hamiltonian plus H_{hf} .

(a) **(5 pts)** Show that H_{hf} splits the ground state into two levels:

$$E_+ = -Ry + \frac{\hbar^2 A}{4}, \quad E_- = -Ry - \frac{3\hbar^2 A}{4}$$

and that corresponding states are triplets and singlet, respectively.

(b) **(20 pts)** Try to estimate the frequency of the emitted radiation as the atom jumps from the triplet to the singlet. To do so, you may assume that the electron and proton are two dipoles μ_e and μ_p separated by a distance a_0 , with an interaction energy of the order

$$\mathcal{H}_{hf} \approx \frac{\mu_e \cdot \mu_p}{a_0^3}$$

Show that this implies that the constant in Eq. (15.1.22) is

$$A \sim \frac{2e}{2mc} \frac{(5.6)e}{2Mc} \frac{1}{a_0^3} \text{ (where 5.6 is the } g \text{ factor for the proton), and}$$

that $\Delta E = E_+ - E_- = A\hbar^2$ is a correction of order $(m/M)a^2$ relative to the ground-state energy. Estimate the frequency of emitted radiation is a few tens of centimeters, using the mnemonics from Ch. 13. The measured value is 21.4 cm. This radiation, called the *21-cm line*, is a way to detect hydrogen in other parts of the universe.

- (c) **(5 pts)** Estimate the probability ratio $P(\text{triplet})/P(\text{singlet})$ of hydrogen atoms in thermal equilibrium at room temperature ($T=300$ K) and show it is $P(\text{triplet})/P(\text{singlet}) \approx 3e^{-\Delta E/kT} \approx 3$

- 3) **(30 pts)** Find the Clebsh-Gordan (CG) coefficients of

(a) **(15 pts)** $\frac{1}{2} \otimes 1 = \frac{3}{2} \oplus \frac{1}{2}$

(b) **(15 pts)** $1 \otimes 1 = 2 \oplus 1 \oplus 0$

- 4) **(10 pts)** Derive Eqs. (15.2.19) and (15.2.20)